

**AMENDMENT NO. 6 SEPTEMBER 2008
TO
IS 15111 (PART 2) : 2002 SELF BALLASTED LAMPS
FOR GENERAL LIGHTING SERVICES
PART 2 PERFORMANCE REQUIREMENTS**

(Page 1, clause 2) — Insert the following at the end:

<i>IS No.</i>	<i>Title</i>
14700 (Part 3/Sec 2) : 1999	Electromagnetic Compatibility (EMC) : Part 3 Limits, Section 2 Limits for Harmonic Current Emission (Equipment input Current ≤ 16 A per phase)

(Page 3, clause 4.5) — Insert the following at the end:

‘4.6 Test for Emission (Radiated and Conducted) of Radio Frequency Disturbances

The number of samples for this test shall be one.’

(Page 3, clause 5.4, line 3) — Substitute ‘requirements of ‘8, 11, 16 and 18 of this standard.’ for ‘requirements of ‘8, 10, 16 and 18 of this standard.’

(Page 3, clause 5.6) — Insert the following at the end:

‘5.7 Emission (Radiated and Conducted) of Radio Frequency Disturbances

Compliance shall be considered to be achieved if the sample selected meets the requirements of Annex C of this standard.

NOTE — For measurement of terminal voltage and transient frequency, see Annex C.’

(Page 4, clause 14) — Substitute the following for the existing matter:

‘14 HARMONICS

14.1 The harmonics of the input current when measured in accordance with IS 14700 (Part 3/Sec 2) shall be as given in 14.1.1 and 14.1.2.

Price Group 2

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14.1.1 For lamps having an active input power greater than 25 W, the harmonic currents shall not exceed the limits given in Table 1A.

Table 1A Limits of Harmonic Current

Harmonic Order	Maximum Permissible Harmonic Current Expressed as a Percentage of the Input of the Fundamental Frequency
<i>n</i>	Percent
(1)	(2)
2	2
3	30 λ ¹⁾
5	10
7	7
9	5
11 ≤ <i>n</i> ≤ 39 (odd harmonic only)	3

¹⁾ λ is the circuit power factor.

14.1.2 For lamps having an active input power smaller than or equal to 25 W, the harmonic current shall comply with one of the following two sets of requirements:

- The harmonic currents shall not exceed the power related limits of col 2 of Table 1B, or;
- The third harmonic current, expressed as a percentage of the fundamental current, shall not exceed 86 percent and the fifth shall not exceed 61 percent. Moreover, the waveform of the input current shall be such that it begins to flow before or at 60°, has its last peak (if there are several peaks per half period) before or at 65° and does not stop flowing before 90°, where the zero crossing of the fundamental supply voltage is assumed to be at 0°.

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Table 1B Limits of Harmonic Current

Harmonic Order <i>n</i>	Maximum Permissible Harmonic Current per Watt mA/W	Maximum Permissible Harmonic Current A
(1)	(2)	(3)
3	2.4	2.30
5	1.9	1.14
7	1.0	0.77
9	0.5	0.40
11	0.35	0.33
$13 \leq n \leq 39$ (odd harmonics only)	$\frac{3.85}{n}$	$0.15 \frac{15}{n}$

[Page 4, clause 16 (see also Amendment No. 3)] — Substitute the following for the existing text:

‘The power factor of the combination of the ballast and the lamp shall be 0.5, *Min.*

NOTE — The value of the power factor shall be 0.85, *Min* with effect from 01-01-2009.’

[Page 4, clause 17.1.3(k)] — Insert the following at the end:

‘m) Emission (radiated and conducted) of radio frequency disturbances.’

(Page 5, Annex A, clause A-2, last line) — Substitute ‘value’ for ‘valve’.

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(Page 7, Annex B) — Insert the following new Annex at the end:

ANNEX C
(Clause 5.7)

**EMISSION (RADIATED AND CONDUCTED) OF
RADIO FREQUENCY DISTURBANCES**

Self ballasted lamps shall comply with the terminal voltage limits given in Table C-1.

Table C-1 Limits of Frequency Range Against Emission

Frequency Range (1)	Limits dB (μ V) ¹⁾	
	Quasi-peak (2)	Average (3)
9 kHz to 50 kHz ²⁾	110	-
50 kHz to 150 kHz ²⁾	90 to 80 ³⁾	-
150 kHz to 0.5 MHz	66 to 56 ³⁾	56 to 46 ³⁾
0.5 MHz to 2.51 MHz	56	46
2.51 MHz to 3.0 MHz	73	63
3.0 MHz to 5.0 MHz	56	46
5.0 MHz to 30 MHz	60	50

¹⁾ At the transition frequency, the lower limit applies.

²⁾ The limit values in the frequency range 9 kHz to 150 kHz are considered to be provisional which may be modified after some years of experience.

³⁾ The limit decreases linearly with the logarithm of the frequency range 9 kHz to 150 kHz and 150 kHz to 0.5 MHz.

Where the light source is operated at a frequency exceeding 100 Hz, the lamp shall comply with the field strength limits given in Table C-2.

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Table C-2 Limits of Loop Diameter Against Transient Frequency Range

Frequency Range		Limits for Loop Diameter dB (μ A) ¹⁾		
MHz	2 m	3 m	4 m	
(1)	(2)	(3)	(4)	
9 kHz to 70 kHz	88	81	75	
70 kHz to 150 kHz	88 to 58 ²⁾	81 to 51 ²⁾	75 to 45 ²⁾	
150 kHz to 2.2 MHz	58 to 26 ²⁾	51 to 22 ²⁾	45 to 16 ²⁾	
2.2 MHz to 3.0 MHz	58	51	45	
3.0 MHz to 30 MHz	22	15 to 16 ³⁾	9 to 12 ³⁾	
¹⁾ At the transition frequency, the lower limit applies.				
²⁾ Decreasing linearity with the logarithm of the frequency.				
³⁾ Increasing linearity with the logarithm of the frequency.				

(Pages 8, 9 and 10, Annex B) — Substitute the following graphs of Fig. 1, Fig. 2 and Fig. 3 for the existing:

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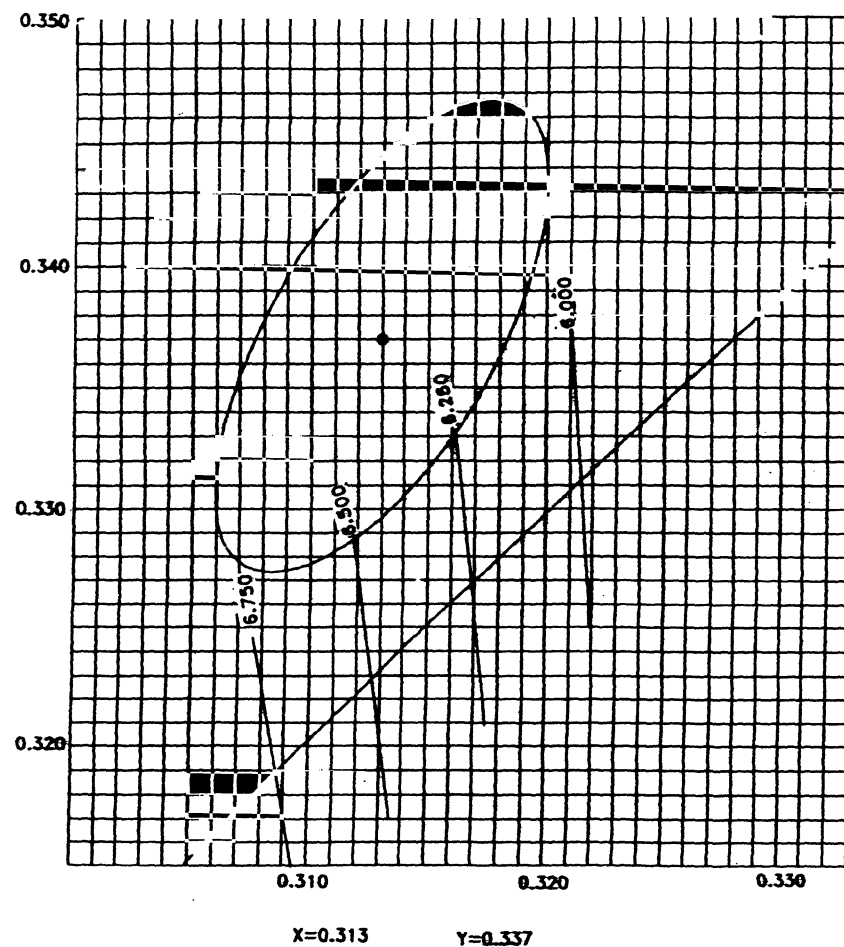


FIG. 1 TOLERANCE AREA FOR STANDARD 'COLOUR' 6500 k

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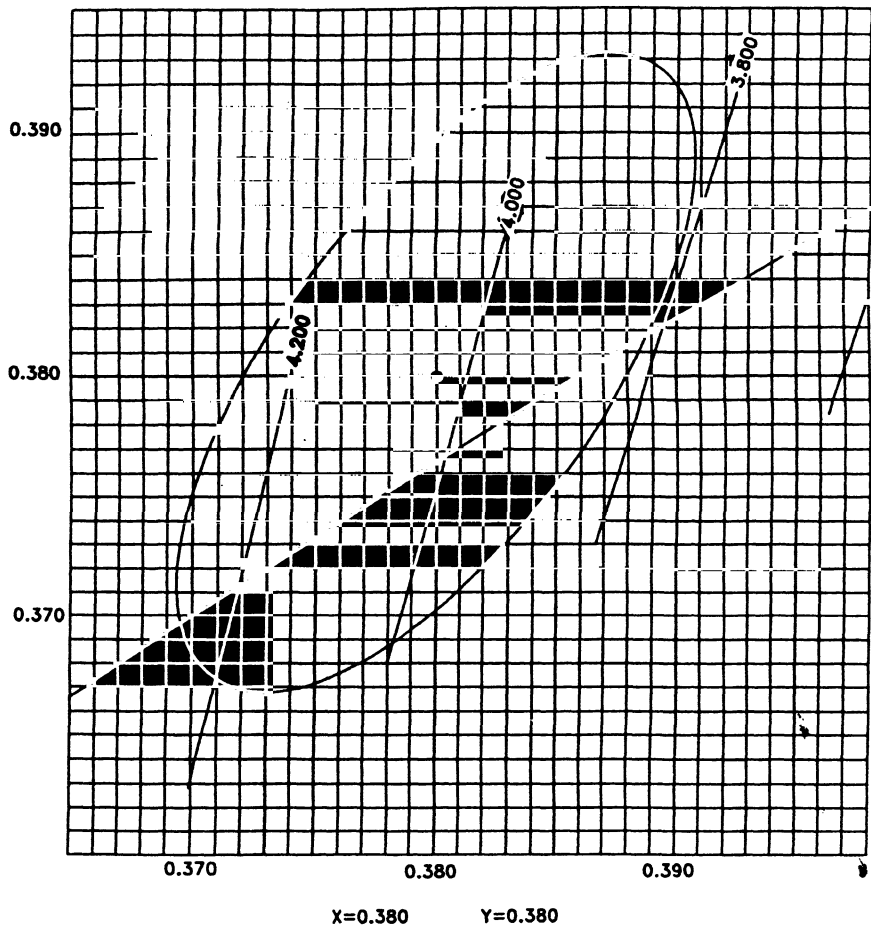


FIG. 2 TOLERANCE AREA FOR STANDARD 'COLOUR' 4000 k

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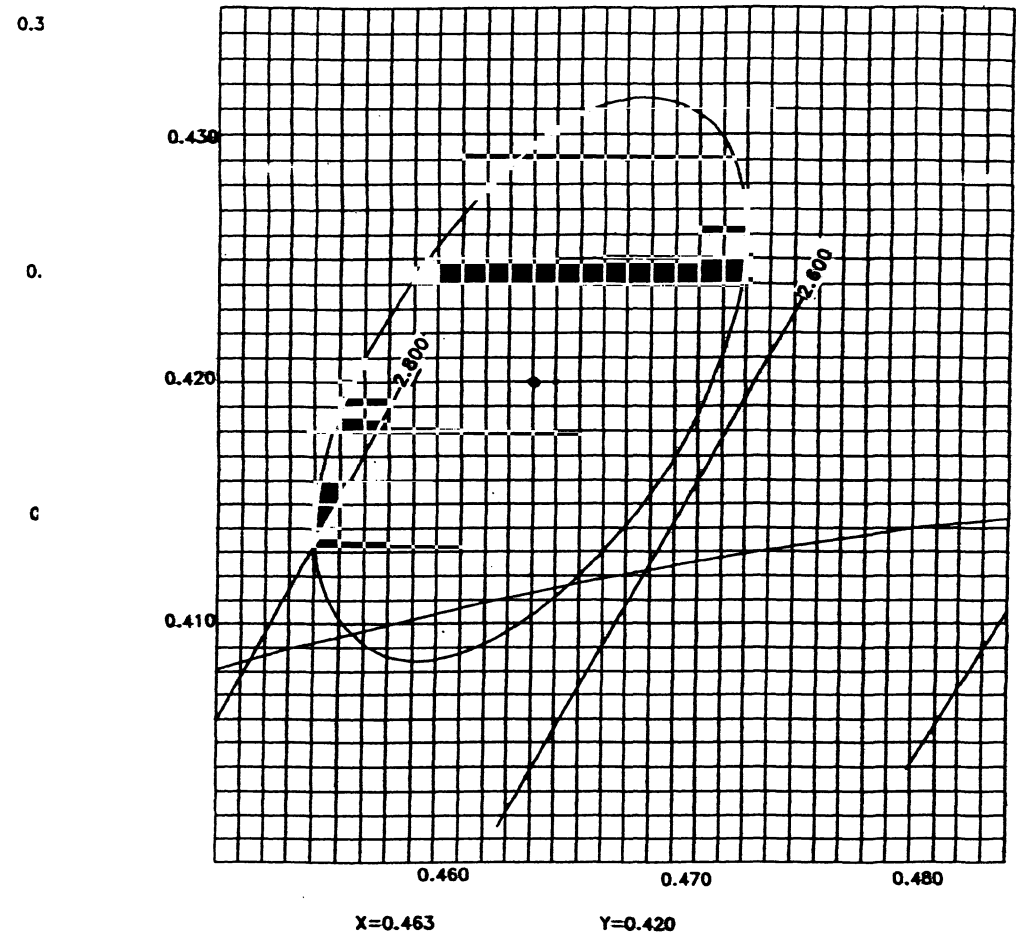


FIG. 3 TOLERANCE AREA FOR STANDARD 'COLOUR' 2700 k

(ET 23)